

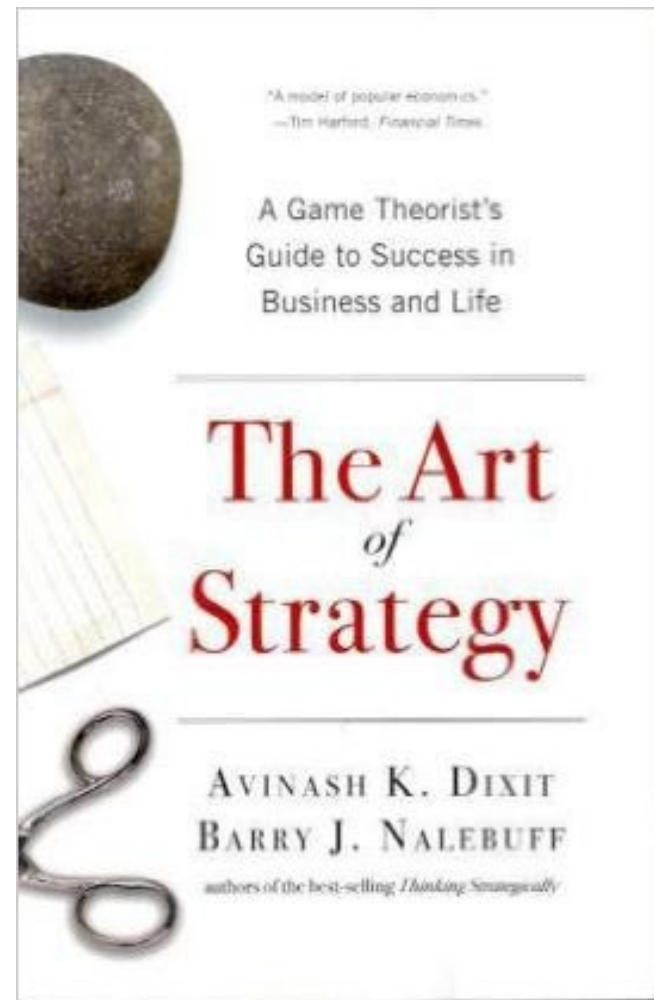
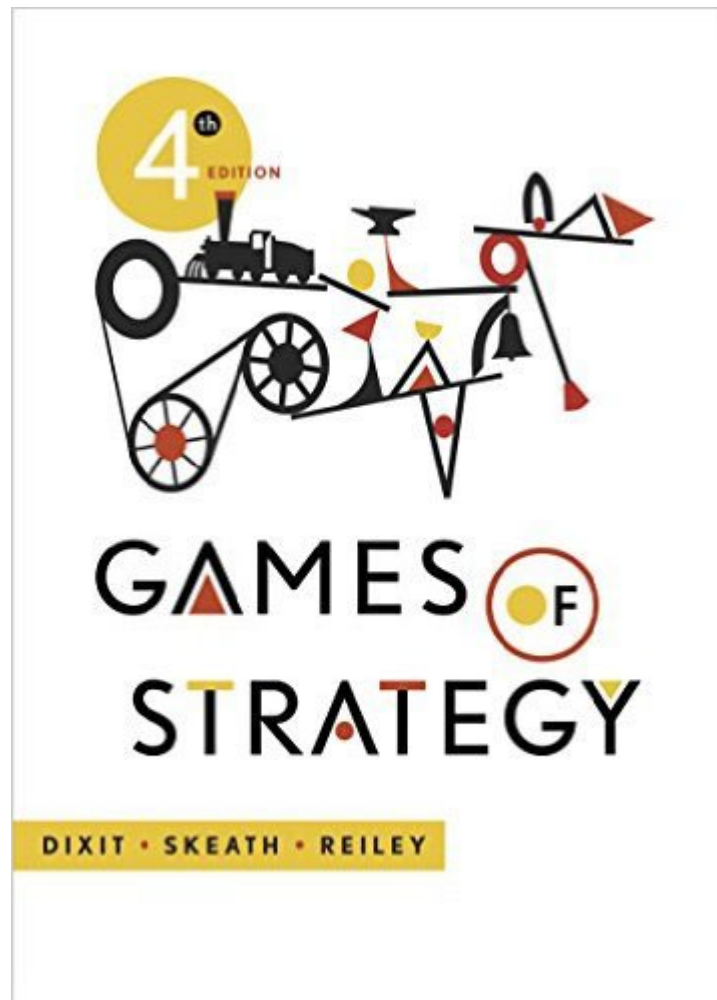
GAME THEORY

Introduce myself ...

- Jidong Zhou (周纪冬); Professor of Economics
 - B.A. from Peking University; Ph.D. from UCL
 - NYU Stern (2011-15); Yale SOM (2015-now)
- Research: applications of game theory, industrial organization, information economics, and behavioral economics
 - Firm competition
 - Digital economy: platform, data, and privacy
 - Market and information design
 - Antitrust and consumer protection
- Teaching: *Game Theory* in the core; *Game Theory & Market Design*; *Games & Information*

Course Materials

- **In class:**
 - *Slides*: basic concepts, examples, and in-class practice questions
- **Outside of class:**
 - *Textbook* (not required): *Games of Strategy* by Dixit, Skeath, and Reiley
 - *Bedtime reading*: *The Art of Strategy* by Dixit and Nalebuff
 - *Canvas*: all the materials are posted in the folder of *Game Theory*
- **Deliverables:**
 - *Problem sets*: one game theory question in PS3; an extra problem set on Canvas (for your own practice)
 - *Final exam*: together with *Basics of Economics* (25%)



Classroom Expectations

- Please do not arrive late or leave early (unless you notify me in advance)
- Please do not start packing up before class ends (unless I go over time)
- Set the right expectation:
 - Course designed for those who haven't learned game theory
 - Questions in the extra problem set are slightly harder than the examples covered in the class

Help

- Jidong Zhou
 - E-Mail: jidong.zhou@yale.edu
 - Office hours: Thursdays (#3528; 4:30pm-5:30pm), or by appointment
- TAs: the same as *Basics of Economics*

Why Game Theory?

- [Needed in many of your future courses at SOM]
- *Game theory* provides a conceptual tool for studying decision-making in environments with *strategic interactions*
- For example, there are only a few major competitors in many markets
 - AT&T, Verizon, T-Mobile; Airbus vs Boeing; Uber vs Lyft; AWS, Azure, GCP
- In such markets, *a firm's profit depends not only on its own actions but also on its rivals'* --- this interdependence is called "*strategic interactions*"
 - This is very different from the case of monopoly or perfect competition
 - *Strategic thinking* is thus important: when making your own decision, you need to consider your competitors' decisions as well

Why Game Theory?

- Many other business scenarios with *strategic interactions*
 - Teamwork, employer-employee relations, upstream-downstream business, auctions, government-business relations, etc.
- Game theory has become a standard tool in economics
 - More than fifteen game theory Nobelists in economics
- It also has numerous applications beyond business and economics
 - Politics, international relations, military, sociology, sports, biology, health care, sustainability, etc.

What They Say in Business ...

I think it is instructive to use game theory analysis... Game theory forces you to see a business situation over many periods from two perspectives: yours and your competitor's.

— Judy Lewent, CFO, Merck

Managers have much to learn from game theory -- provided they use it to clarify their thinking, not as a substitute for business experience.

— The Economist

Game theory is hot.

— The Wall Street Journal

Game: Guess 2/3 of the Average

- Each of you guesses an integer number between **1** and **100**
- The winner is the person whose guess is closest to **2/3 of the average**

Prize: \$10

[by Venmo]

- MW 10:10am (**Blue**): **m3n4bax34**
- MW 2:40pm (**Silver**): **kgp6bax34**
- TTh 10:10am (**Green**): **82u7bax34**
- TTh 1:00pm (**Gold**): **m8r8bax34**
- TTh 2:40pm (**Red**): **3pn9bax34**

A Running Example: Software Launch Game

- You own a software company and are deciding whether to launch a new software package. Your rival faces a similar decision.
 - If both companies launch, each loses \$20 million
 - If you launch and your rival does not, you earn \$30 million and your rival loses \$10 million; vice versa
 - If neither company launches, stay with the status quo
- Two possible scenarios:
 - *Sequential* moves: you make your decision first, and then your rival makes their decision after observing yours
 - *Simultaneous* moves: you and your rival make decisions simultaneously*

*Don't take the word "simultaneously" too literally. It just means that when you and your rival make decisions, you two don't observe each other's choices.

Basic Elements and Assumptions

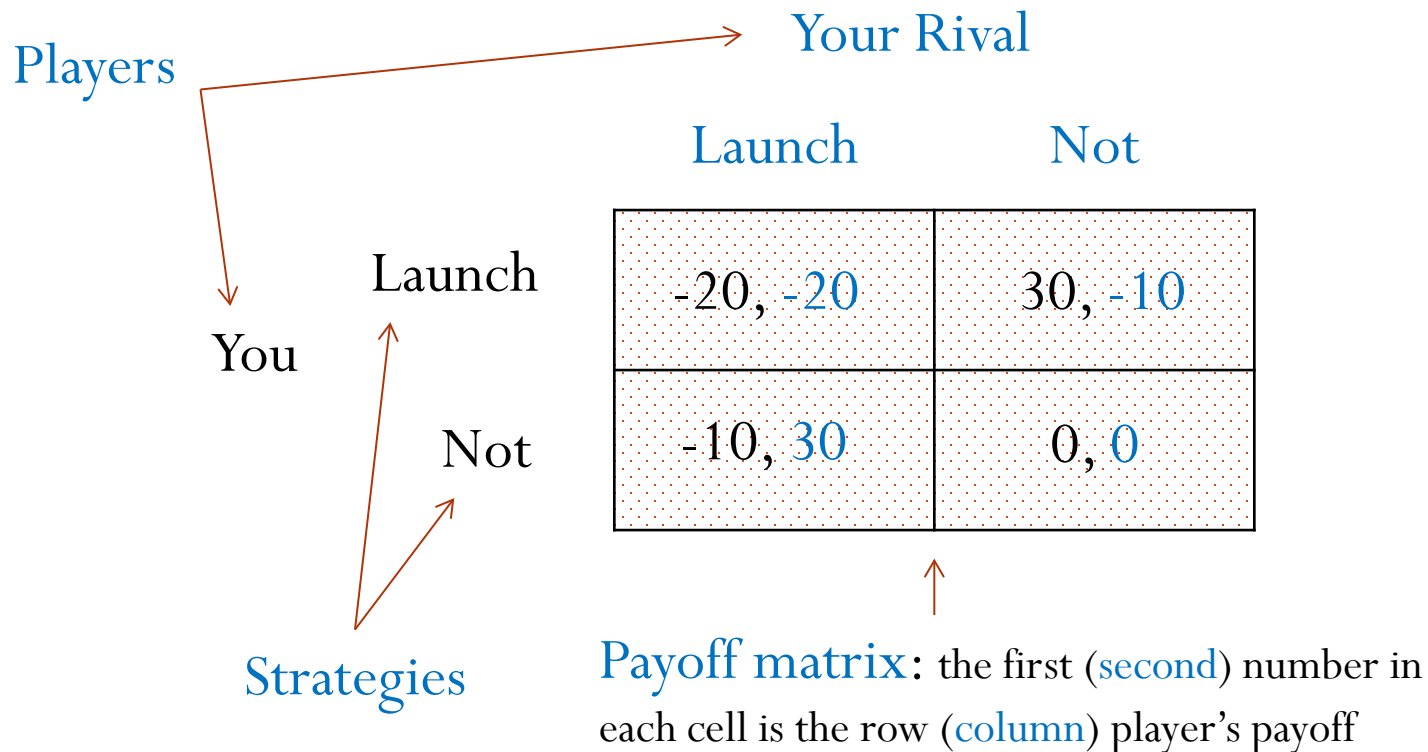
- **Basic elements** of a game:
 - *Players*: you and your rival in this example
 - *Strategies / actions*: launch or not
 - *Payoffs*: profits at each possible outcome
 - *Rules*: sequential or simultaneous move
- **Assumption**: *each player is “**rational**,” aiming to maximize their **own** payoffs; this is commonly known among players*
 - In this example payoffs are assumed to be profits (*NOT winning or losing*)
 - *More generally, payoffs can capture everything players care about* (e.g., revenues, market shares, relative performance, fairness, other players' welfare, social responsibility, etc.)
- Game theory predicts / advises *how each player will / should act in a game*

Outline

- How to describe a game?
- Simultaneous-move games:
 - How to solve it (or to make a prediction)?
 - Important simultaneous-move games [Class 2]
- Sequential-move games: [Class 3]
 - How to solve it?
 - Important sequential-move games
- Applications to price/quantity competition [Class 4]

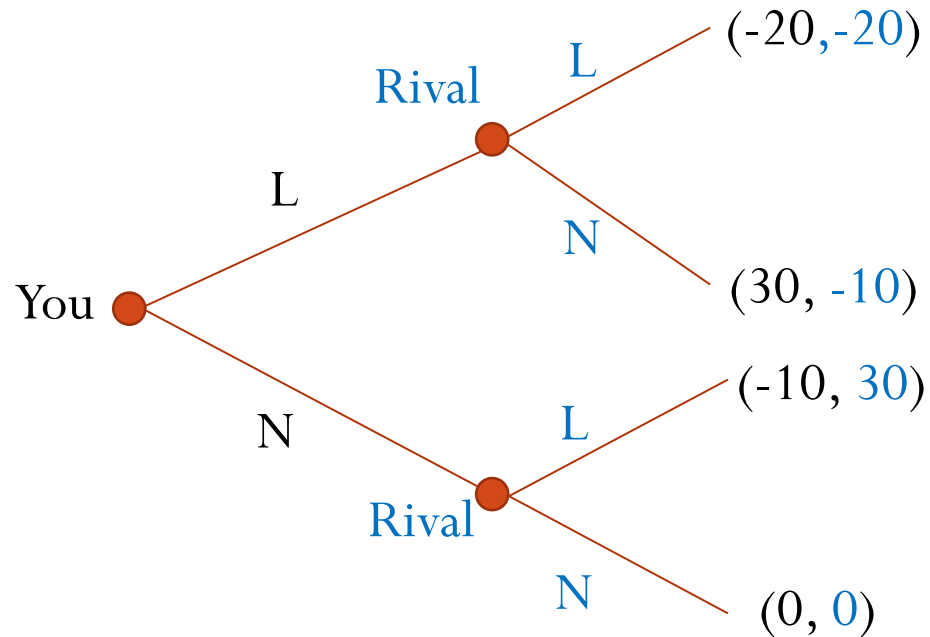
Represent a Simultaneous-Move Game

- **Matrix form** (or normal form)



Represent a Sequential-Move Game

- **Game tree** (or extensive form)



In this mini course, we assume all the information in the matrix or game tree is *publicly* known among players.

Simultaneous-Move Games

Dominant Strategy

- **Definition.** A strategy is called a player's *dominant strategy* if it earns the player a greater payoff than *all other* strategies the player can use, *regardless of other players' choices*
 - If a player has a dominant strategy, she should choose it

		Player II	
		L	R
Player I	T	0, <u>7</u>	4, 5
	B	1, <u>4</u>	2, 3

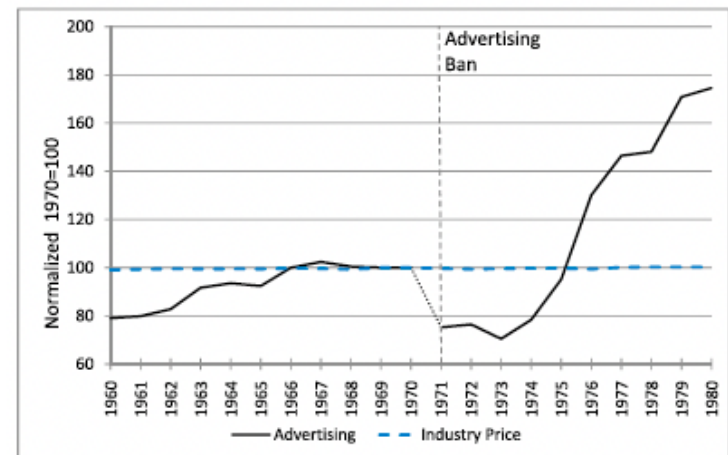
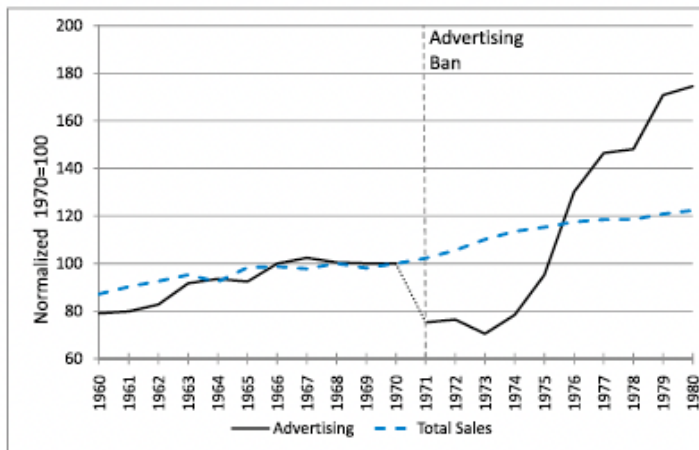
L is player II's dominant strategy

Remark. Player 1's maximal payoff is 4, but it is impossible to achieve due to the "constraint" from player 2's rational choice; similar for player 2

Prediction is (B,L)

Advertising competition and regulation

- Cigarette advertising on TV and radio was banned in 1971
- The black curves below depict how industry ad spending changed over time
- In the short-medium run, did this regulation harm or improve industry profit?
- Why didn't companies spontaneously cut ad spending before regulation?



A Toy Model of Advertising Competition

- Two competitors: Reynolds, and Philip Morris
 - Decide simultaneously whether to advertise heavily or lightly
 - Heavy advertising costs \$30 m, while light advertising costs only \$10 m
- The cigarette market has a steady market value of \$100 m
 - If the two companies choose the same ad strategy, they split the market, each getting \$50 m
 - If they choose different strategies, the one with more advertising obtains a larger market share and earns \$80 m, and the other earns only \$20m
- Suppose each firm aims to maximize their own profit (=market value – ad cost). What would they do *before* the regulation?

Advertising Competition

		Philip Morris	
		H Ad	L Ad
Reynolds	H Ad	20, 20	50, 10
	L Ad	10, 50	40, 40

“H Ad” is each firm’s
dominant strategy

- Both advertise heavily and each earns 20, though light advertising is a *collectively better* outcome
- Conflict between *individual rationality* and *collective rationality*
- Under regulation, each firm is forced to choose light advertising, profit rising to 40---**regulation helps firms escape the dilemma!**

One more exercise

- Does any player have a dominant strategy? Prediction?

		Player II		
		L	C	R
Player I	T	9, 5	5, 6	1, <u>7</u>
	M	1, 3	2, 4	3, <u>5</u>
	B	2, 7	3, 6	2, <u>8</u>

R is player II's
dominant strategy

Prediction is (M,R)

Dominated Strategy

- **Definition.** A strategy is called a player's *dominated strategy* if it earns the player a lower payoff than *some* other strategy the player can use, *regardless of other players' choices*
 - A player should never adopt dominated strategies

		Player II		
		L	C	R
Player I	T	9, 3	8, 6	1, 4
	B	1, 7	2, 5	3, 3

For player II, R is dominated by C

Prediction is (T,C)

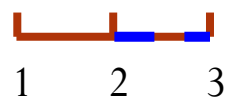
Example: The Guessing Game

- Recall the game of guessing $2/3$ of the average
- Impossible to draw a matrix ...
- But we can still use the concept of dominated strategy
 - Does each player have any dominated strategies?
 - What's the prediction?









$$\frac{4}{3}$$


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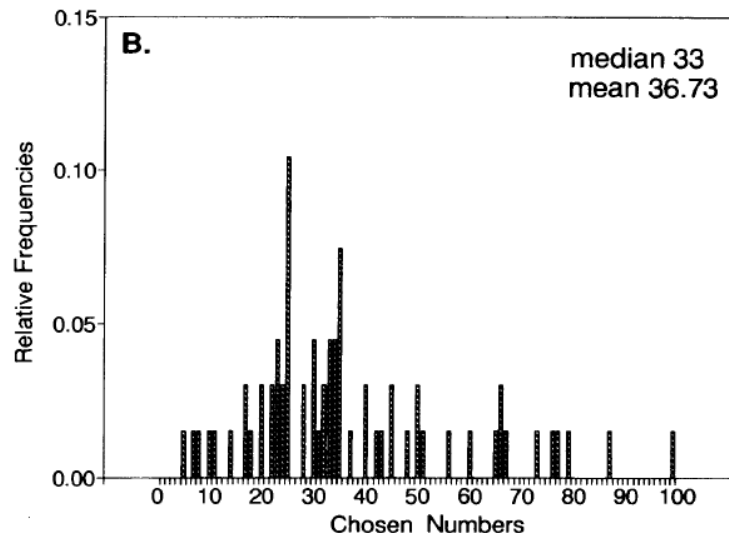
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Example: The Guessing Game

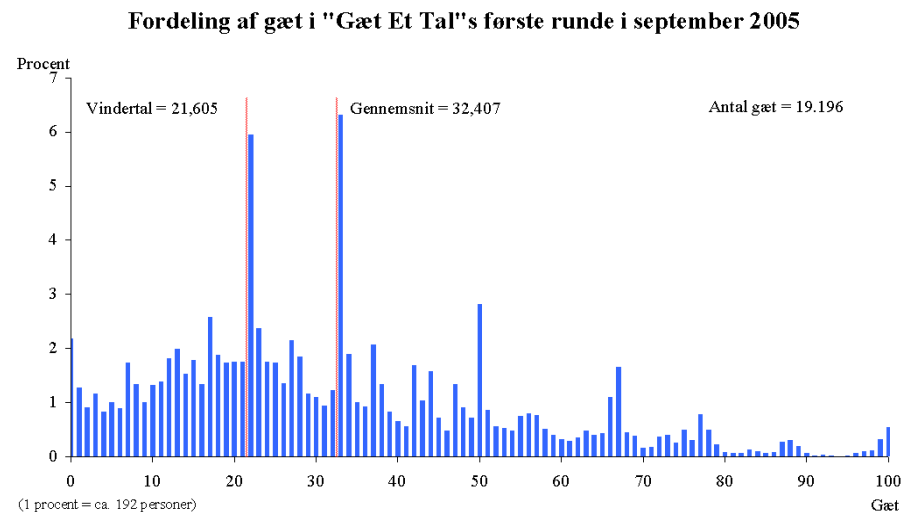
- Recall the game of guessing $2/3$ of the average
- Impossible to draw a matrix ...
- But we can still use the concept of dominated strategy
 - Does each player have any dominated strategies?
 - What's the prediction?
 - By **repeatedly eliminating** dominated strategies, we predict that **every player guesses 1**
 - Does that fit reality well?

Real-world evidence

- People behave very differently from the theory if they play the game **only once**
 - Many people think only **one or two steps** ahead
 - This remains true even if the prize is much larger
- *Strategic thinking* also requires sound judgment of other players' sophistication

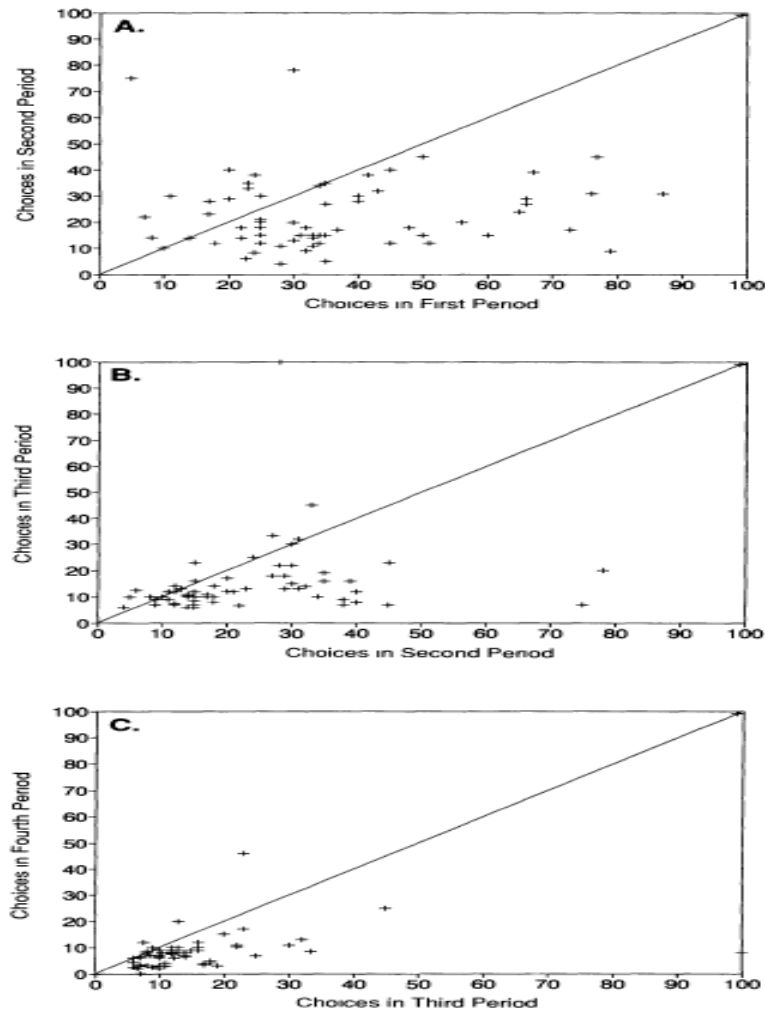


Lab experimental evidence from
Nagel (1995) in *American Economic Review*



Field experimental evidence from Denmark
with a prize of 5,000 kroner

- However, in **repeated** play with learning opportunities, players behave more as theory predicts



Homework:

- 0 is included?
- $2/3$ becomes 1?

Lab experimental evidence from Nagel (1995) in *American Economic Review*

Second-Price Auctions

[for you own reading; see also the class note]

- **Second-price auctions (SPA):**
 - Bidders compete for an object by submitting their bids simultaneously
 - The highest bid wins, and the winner pays *the second highest bid* (or the highest losing bid)
 - E.g. Google's keyword auction is a variant of SPA
- You know your own value for the object (say, \$10), but you have no idea of other bidders' values
- How will you bid?
- ***Bidding at your own value is a dominant strategy!***
 - Only the *highest* bid among *other* bidders (denoted by b) matters for you
 - b can take many possible values, but must be in one of the *two categories* below
 - Poss. 1: b is above your value. Can you do better than bidding at your value?
 - Poss. 2: b is below your value. Can you do better than bidding at your value?

Summary so far ...

- Game theory studies decision making with strategic interactions
 - Think strategically: put yourself into other players' shoes
- A rational player should adopt a dominant strategy if any
 - However, individual rationality can lead to collective irrationality
- A rational player should avoid using dominated strategies